MIL-A-60146A(MU)
29 July 1966
SUPERSEDING
MIL-A-60146(MU)
18 January 1965

MILITARY SPECIFICATION

AMPLIFIER, ELECTRONIC CONTROL: 10527247

1. SCOPE

1.1 This specification covers one type of electronic control amplifier, part of Antioscillation Sight System XM58 employed in a UH-1B helicopter (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-T-152	Treatment, Moisture and Fungus-Resistant, of Communication, Electronic and Associated Electrical Equipment
MIL-I-6181	Interference Control Requirements, Aircraft Equipment
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Materiel, Packaging of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions

DRAWINGS

U.S. Army, Frankford Arsenal

F10527247	Amplifier, Electronic Control

FSC 1270

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PACKAGING DATA SHEET

MIL-P-14232/P10527247 Packaging of Amplifier, Electronic Control: 10527247

(Copies of specifications, standards, drawings, and packaging data sheets required by supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article testing.- Requirements for the submission of first article samples by the contractor for testing shall be as specified im the contract.

3.2 Fabrication.- The electronic control amplifier shall be fabricated in accordance with Drawing F10527247 and drawings pertaining thereto.

3.2.1 Treatment, moisture and fungus-resistant.- The electronic control amplifier shall be treated for moisture and fungus resistance in accordance with MIL-T-152.

3.3 General specification.- The contractor shall be responsible for adherence to the following requirements of MIL-F-13926:

(a) Dimensions and tolerances
(b) Inorganic protective surface finishes
(c) Part identification and marking
(d) Workmanship
(e) Soldering electrical connections

3.4 Environmental.

3.4.1 Storage temperature.- The electronic control amplifier shall meet the performance requirements of 3.5 at 60 to 90° Fahrenheit (F) after being subjected to temperatures at -80° and at +160°F.

3.4.2 Operating temperature.- The electronic control amplifier shall meet the performance requirements of 3.5 while exposed to temperatures at -40° and at +150°F.

3.4.3 Altitude.- The electronic control amplifier shall meet the performance requirements of 3.5 at an altitude of 30,000 feet (ft) above sea level.

3.4.4 Shock.- The electronic control amplifier shall withstand impact shocks up to 15 G's, each shock impulse having a time duration of 11 $\pm$ 1 millisecond (ms). Subsequently the electronic control amplifier shall meet the performance requirement of 3.5.

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3.4.5 Vibration.- The electronic control amplifier shall meet the performance requirements of 3.5 after being subjected to vibrations up to 5 G's over the frequency range of 5 to 500 cycles per second (cps).

3.5 Performance.

3.5.1 Input.- The electronic control amplifier shall accept an input voltage from 25 to 31 volts dc (Vdc). Nominal input shall be 28 Vdc.

3.5.2 Outputs.- The electronic control amplifier shall provide the following outputs:

(a) A 26-volt, 400 cps, sinusoidal output to drive a maximum load of 14 watts, resistive for all specified input variations.

(b) Power capability of 40 watts at approximately 20 Vdc for each of two control motor loads.

(c) A 400 cps sinusoidal gyro pickoff excitation output to drive a maximum load of 86.0 ohms.

3.5.3 Voltage regulation.- The voltage regulation of 26-volt, 400 cps output shall be 26 volts rms ± 7.5% for input variations specified in 3.5.1 and output load variations of 7 to 14 watts.

3.5.4 Frequency regulation.- The frequency regulation of the 26V output shall be ± 2.5% for specified input variations and for one-half to full-rated power output from the 26V source.

3.5.5 Harmonic distortion.- The harmonic distortion of the 26V, 400 cps output shall be less than 8% at the specified rated load.

3.5.6 D-C (control motor) outputs.

3.5.6.1 Azimuth axis.- The dc output labeled azimuth Control Motor Signal shall be dc with ripple less than 3 volts peak to peak at predominantly 800 cps for any unsaturated magnitude up to ± 19 volts dc and with a nominal 10 ohm resistive load.

3.5.6.2 Elevation axis.- The dc output labeled Elevation Control Motor Signal shall be dc with ripple less than 8 volts rms at predominantly 800 cps for any unsaturated output. The output shall have a linearity of ± 20% between minus 10 and plus 10 volts dc. The gain at plus 10 volts shall be 30 ± 20% with a nominal 10 ohm resistor connected across the motor control output.

3.5.7 D-C output sensing.- The sense of the control motor signal outputs shall be positive with respect to a-c input signals. An a-c input in phase with the 26-volt, 400 cps output shall produce a positive d-c voltage at the control motor signal outputs.

3.5.8 Gyro pickoff excitation output.- The gyro pickoff excitation output shall be 80 ± 3 percent of the 400 cps output voltage when loaded with a resistive load of 86 ohms.

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3.5.9 Time delay.- The control motor voltages shall not be applied to their respective loads until after a time delay following the application of primary 28V power. This time delay shall be from 15 seconds to 40 seconds.

3.5.10 Step response.- The response of azimuth and elevation control motor output signals shall conform to the characteristics of Table 1 following the application of a step input signal at the gyro pickoff signal input. The characteristics are defined by Figure 1.

TABLE I

CONTROL MOTOR RESPONSE CHARACTERISTIC

	Requirement		
Servo Channel	1 Azimuth	2 Azimuth	3 Elevation
Input (400 cps, ac, 0° phase)	150 mv	30 mv	50 mv
Output characteristics (Figure 1			
V_1	6.5V ± 15%		1.5V ± 20%
V_2		13.5V ± 15%	Greater than 2.0V
T_1 (less than)	0.01 sec		0.01 sec
T_2 (± 20%)		1.3 sec	
S^2 (volts/sec ± 15%)		18	
Curve	a	b	c

3.6 Radio interference suppression.- Interference control requirements shall be in accordance with MIL-I-6181.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article testing.- When the contract specifies testing of preproduction or initial production samples, the following paragraphs shall apply as applicable.

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4.2.1 Preproduction test.- Preproduction test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, two preproduction samples shall be submitted to the procuring agency for approval of design and construction features. Preproduction samples shall meet all inspection provisions of 4.3, 4.4 and 4.6 as well as any additional tests that may be specified in the contract.

4.2.2 Initial production test.- Initial production test samples shall be submitted in accordance with contract requirements. Unless otherwise specified in the contract, samples shall be the first two production units and shall have been manufactured using the same materials, processes and procedures as will be used in regular production. Initial production testing shall be in accordance with 4.3, 4.4 and 4.6 and shall be witnessed by the Government representative for all examinations and tests. The Government reserves the right to conduct any or all examinations or tests as may be deemed necessary.

4.2.3 First article disapproval.- Failure of any one of a first article sample submitted shall be cause for disapproval by the Government and subsequent action as required by the contract clauses applicable to first article testing.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the Contracting Officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the List of Inspection Documents (IEL), when provided with the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 First article tests.- Unless otherwise specified, the requirements and tests of Table II shall be conducted on first article tests only. In the event that the processes, materials or other changes in design or manufacture occur subsequent to first article approval, the first items so produced shall be submitted to the Government in accordance with 4.2 for approval prior to resumption of production.

TABLE II

	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Altitude	3.4.3	4.5.2.3
302.	Shock	3.4.4	4.6.2.4
303.	Vibration	3.4.5	4.6.2.5
304.	Radio interference suppression	3.6	4.6.4

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4.3.2.3 Final acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables III and IV herein. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE III

CLASSIFICATION OF DEFECTS

CRITICAL: NONE MAJOR: AQL 1.5 Percent Defective	REQUIREMENT	TEST PROCEDURE
101. Treatment, Moisture and Fungus-resistance	3.2.1	4.6.4
102. Input	3.5.1	4.6.3.1
103. Output	3.5.2	4.6.3.2
104. Voltage regulation	3.5.3	4.6.3.2
105. Frequency regulation	3.5.4	4.6.3.2
106. Harmonic distortion	3.5.5	4.6.3.2
107. D-C (control motor) outputs	3.5.6	4.6.3.3
108. D-C output sensing	3.5.7	4.6.3.4
109. Gyro pickoff excitation output	3.5.8	4.6.3.5
110. Time delay	3.5.9	4.6.3.6
111. Step response	3.5.10	4.6.3.7
MINOR: NONE		

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.- Three amplifiers shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table IV.

TABLE IV

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
401. Fabrication	3.2	Applicable drawing - visual
402. General specification	3.3	MIL-F-13926 - visual
403. Temperature (storage)	3.4.1	4.6.1.1
404. Temperature (operating)	3.4.2	4.6.1.2

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

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4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.6 Test methods and procedures.

4.6.1 Treatment, moisture and fungus-resistant.- The electronic control amplifier shall be tested in accordance with MIL-T-152 to insure compliance with 3.2.1.

4.6.2 Environmental.

4.6.2.1 Storage temperature.- The electronic control amplifier, while in non-operating condition, shall be subjected to the Temperature Cycling Test, Procedure II of MIL-F-13926 using Conditions A and H. Upon stabilization at 60° to 90°F, the electronic control amplifier shall meet the performance requirements specified in 3.5.

4.6.2.2 Operating temperature.- The electronic control amplifier shall be subjected to the Temperature Cycling Test Procedure I of MIL-F-13926 except that the temperature range shall be -40° to +150°F. During temperature stabilization at -40°F, +150°F and room ambient, the electronic control amplifier shall meet the performance requirements specified in 3.5.

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4.6.2.3 Altitude test.- The electronic control amplifier shall be placed within a test chamber and the absolute internal pressure reduced to 8.88 inches of mercury (corresponding to an altitude of 30,000 ft above sea level). After sufficient time has elapsed to allow all entrapped air to excape, the electronic control amplifier shall meet the requirements specified in 3.5.

4.6.2.4 Shock test.- The electronic control amplifier shall be tested in accordance with MIL-F-13926, Procedure II. The electronic control amplifier shall not suffer damage or subsequently fail to provide the performance specified in 3.5.

4.6.2.5 Vibration test.- With the electronic control amplifier mounted and operating at room ambient conditions, the table shall be vibrated along a horizontal direction with the frequency varying between 5 and 500 cps. For frequencies between 5 and 20 cps, the double amplitude shall be 0.1 inch. For frequencies between 20 and 33 cps, the amplitude shall be varied so as to maintain a 2 G acceleration. For frequencies between 33 and 52 cps, the double amplitude shall be 0.036 inch. For frequencies between 52 and 500 cps, the amplitude shall be varied so as to maintain a 5 G acceleration. The rate of frequency change shall be logarithmic and shall be such that a complete cycle (5-500-5) will consume approximately 15 minutes. The test shall continue for four 15-minute cycles for a total of 1 hour. (When provisions for logarithmic cycling is not available, a linear rate of frequency change may be used.) The cycling test shall be repeated with the axis of vibration changed 90 degrees in the horizontal direction and then with the axis of vibration changed to vertical. At the completion of each direction of vibration, the electronic control amplifier shall be closely inspected for any evidence of mechanical failure, loose or broken parts. Subsequently, the electronic control amplifier shall meet the performance of 3.5.

4.6.3 Performance.

4.6.3.1 Input voltage test.- Using standard measuring equipment, verify that the input voltage to the electronic control amplifier is in compliance with 3.5.1.

4.6.3.2 Regulation test.- Connect standard measuring equipment, power supplies, and dummy loads to the amplifier as shown in Figure 2. With S1, S2, and S3 open, measure the a-c voltage (M4), the frequency and harmonic distortion (M5) of the a-c output (pins F and G) while varying the d-c input at 25, 28, and 31V dc (M1). Repeat the test with S1 closed. The test data shall comply with 3.5.2 for output voltage test, 3.5.3 for voltage regulation, 3.5.4 for frequency regulation, and 3.5.5 for harmonic distortion.

4.6.3.3 D-C control motor output test.- Using the same test setup as for 4.6.3.2 set the d-c power supply to a nominal 28V dc (M1). S1 is closed. Apply a sufficiently large a-c signal via the S2 and S3 switches until approximately + 19V or - 19V in azimuth and +10 or -10 volts in elevation is observed on M7 and M8, the control motor loads. Observe both

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outputs on the oscilloscope and measure the superimposed noise signal. The magnitude of this noise signal shall comply with 3.5.6.

4.6.3.4 D-C output sensing test.- Keeping the test setup of 4.6.3.2 and the test conditions of 4.6.3.3 compare each input signal phase relationship with the 26V, 400 cps output. Use either the oscilloscope or other phase-measuring devices. The phase relationship and the sense of the d-c output shall comply with 3.5.7.

4.6.3.5 Gyro pickoff excitation test.- During the preceding test (4.6.3.4, measure the voltage across the 86-ohm load (pins G and J) with meter M6 for conformance with 3.5.8.

4.6.3.6 Time delay test.- Using the test setup of Figure 2, all voltages at nominal, S1 open, S2 and S3 closed and supplying a non-saturating input voltage, open S4, wait at least 30 seconds then close S4 and measure the time it takes for a voltage reading to show on M7 and M8. Repeat 5 times and average making certain that the on-off cycle is not more frequent for conformance with 3.5.9.

4.6.3.7 Step response test.- Same setup as in Figure 2, nominal 28V dc, S1 and S4 closed, S2 and S3 open. Replace, in turn, M7 and M8 with an oscillographic recorder (SME) and adjust proper speed to obtain an adequate recording of control motor voltage. Apply an appropriate amplitude input signal, first into the azimuth gyro pickoff input (M2) and then into the elevation gyro pickoff input (M3), recording each time the voltage response of the appropriate control motor output. Test procedures are itemized in Table V below with the specified performance with reference to Table I of 3.5.10.

TABLE V

Test No.	Gyro P.O. Input (400 cps 0° phase)	Channel	Recorder Paper Speed	Requirement No. (Table I, 3.5.10)
1	150 mv	Azimuth	100 mm/sec	1
2	30 mv	Azimuth	25 mm/sec	2
3	2.5 V	Elevation	100 mm/sec	3

4.6.4 Radio interference test.- The electronic control amplifier shall be tested in accordance with MIL-I-6181 for conformance with 3.6.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and Packaging Data Sheet MIL-P-14232/P10527247. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Amplifier, Electronic Control: 10527247 contains and combines two amplifiers to provide servo stabilization and

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control of the elevation and azimuth gimbals of the Sight Assembly: 10527236. In addition, it contains power control switching circuitry and feedback stabilization networks for the amplifiers. The electronic control amplifier also combines the Power Supply: 10527250 with various input-output devices to provide regulated power to the servo amplifiers and other components of the Antioscillation Sighting System XM58.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of applicable levels of preservation, packaging, and packing (see 5.1).
(c) Applicable packaging data sheet number (see 5.1).
(d) Applicable stock number.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1270-A038Y

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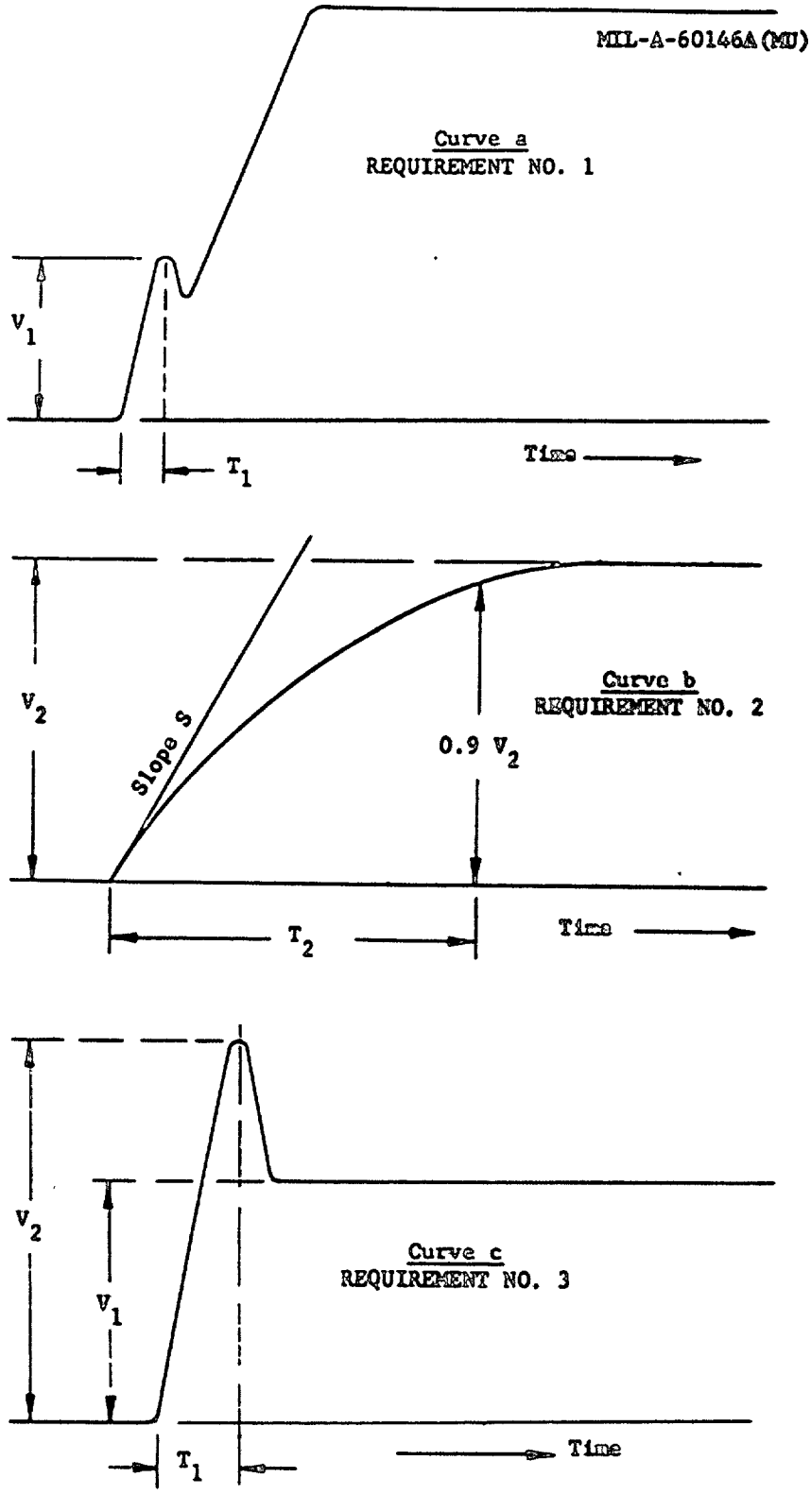


Figure 1: Step Response Waveforms for Azimuth and Elevation Torquer Outputs

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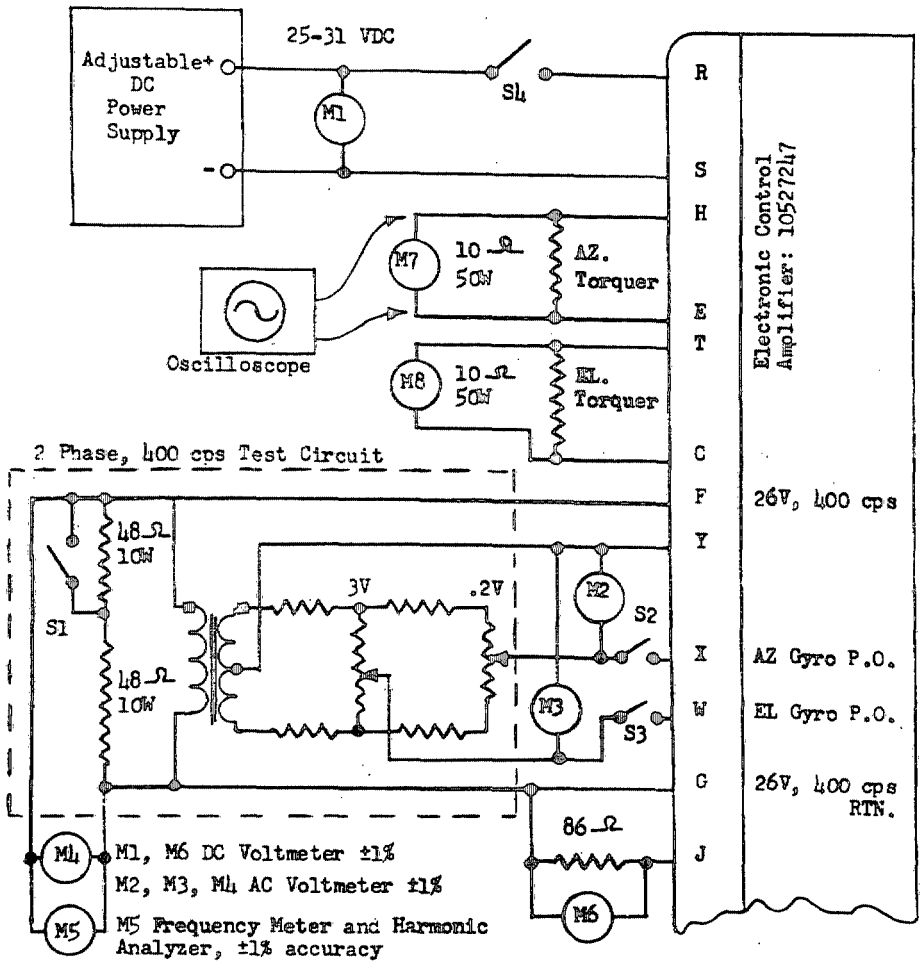


Figure 2. Test Set-Up for Electronic Control Amplifier: 1052724?

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

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